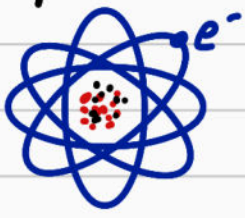
Chapter 6 ③/60

Bohr's Atomic Model | Hydrogen atom spectra | Spectrometer & Types of spectra | X-rays

1 Bohr's atomic model:



He agreed With Rutherford:

① The mass of the atom is concentrated at its center [nucleus] which consists of positive protons & neutral neutrons. → nucleus is positive

② negative charges [negative electrons] revolve around nucleus in standing Waves → Energy levels

electron → particle has wave nature (de-Brogli)

③ the Atom neutral Na_{11} ↔ 11 e^-, 11 p^+

Positive Protons = negative electrons = atomic no.

Then Bohr added:

~~$F = \frac{-Kq}{r^2}$~~ → incl

④ Coloumb's Law (Electric force) & → كولوم

Newton's Law ($\Sigma F = ma$ Mechanical force) are applied on the atom.

(5) The radius of any energy level can be calculated by $2\pi r = n\lambda$

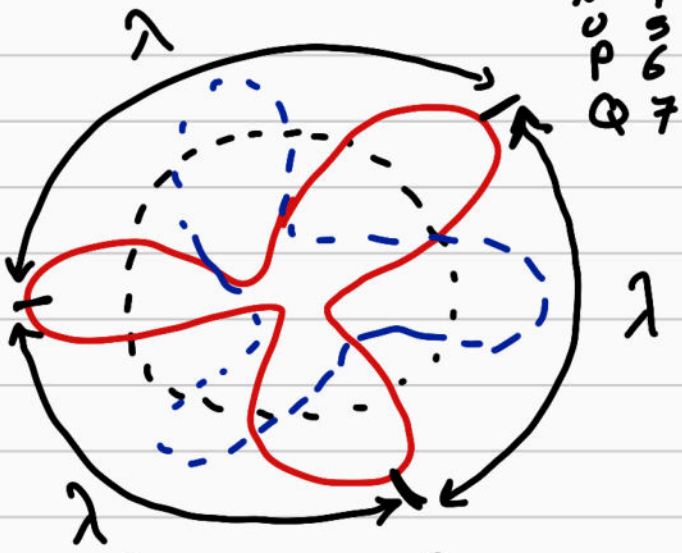
K 1
L 2
M 3
N 4
O 5
P 6
Q 7

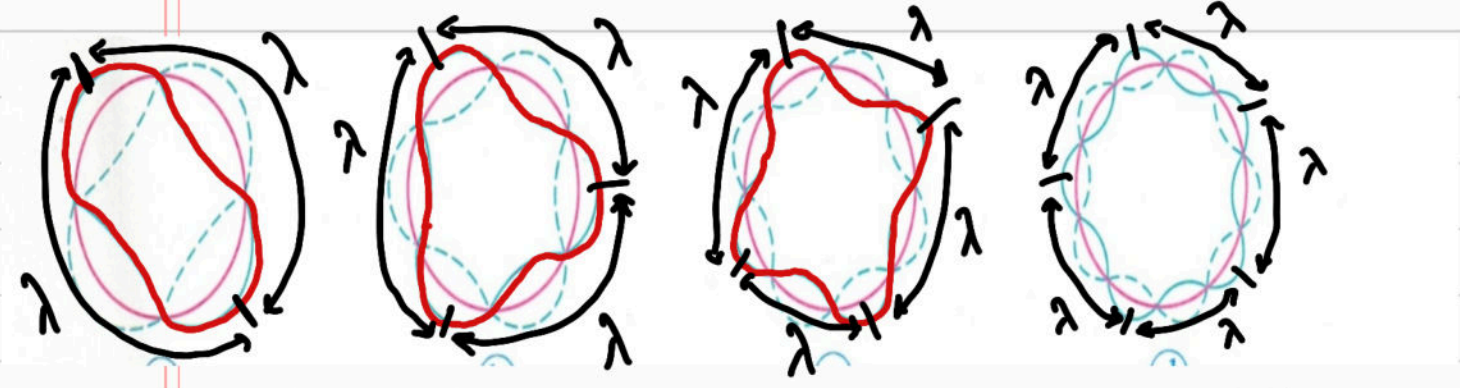
$2\pi r = 3\lambda$

n

energy level

$n=3$ M-Level





$2\lambda = 2\pi r$ $n=2$ L-Level

$3\lambda = 2\pi r$ $n=3$ M-Level

$4\lambda = 2\pi r$ $n=4$ N-Level

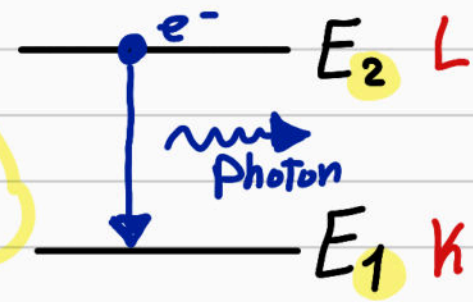
$5\lambda = 2\pi r$ $n=5$ O-Level

$2\pi r = n\lambda \longrightarrow \lambda = \frac{h}{p_L} = \frac{h}{m_e v} \longrightarrow KE = \frac{1}{2} m_e v^2$

$h = 6.625 \times 10^{-34}$ J.s (planck's constant)

$m_e = 9.1 \times 10^{-31}$ kg (mass of electron)

⑥ The energy of photon Can be calculated by $\Delta E_{photon} = E_{higher} - E_{lower}$ (i.e. $E_2 - E_1$)





Absoreb photon
"Excitation"

relase/emit photon
"Relaxation"

$\Delta E = E_1 - E_2$

$= \ominus 10.2eV$

Absoreb

$\Delta E = E_2 - E_1$

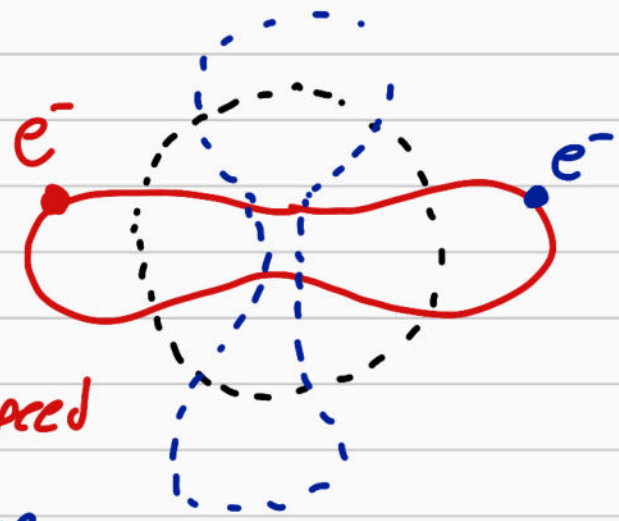
$= \oplus 10.2eV$

emiT/Relase

if Two electrons are in same energy level they have

1. Same energy
2. Same velocity/speed
3. Same λ
4. Same r

} same level

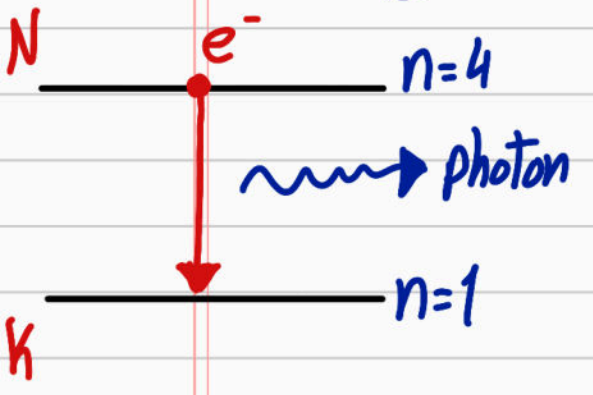


2 Hydrogen Atom Spectrum:

any level in hydrogen atom its energy can be calculated by

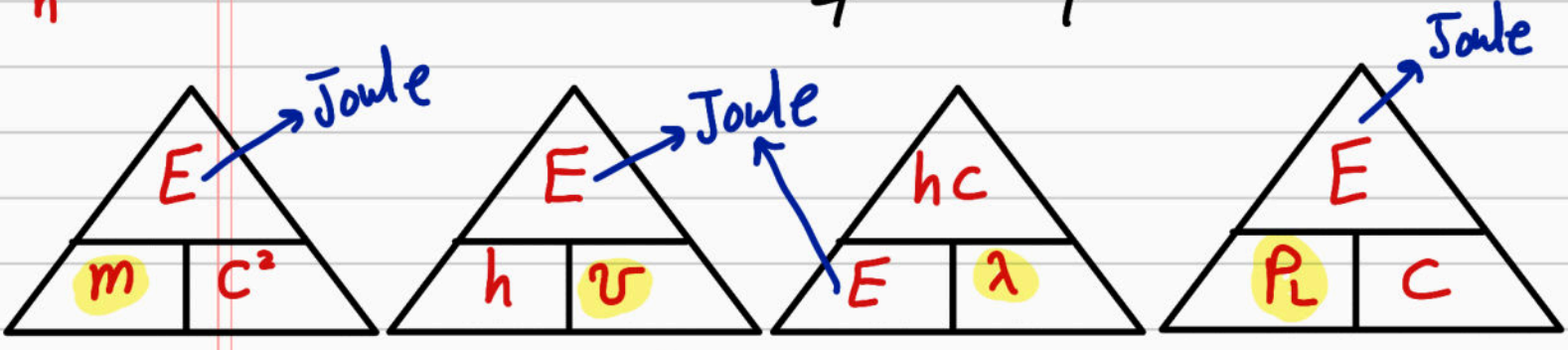
$$E_n = \frac{-13.6}{n^2}\ eV \xrightarrow{\times 1.6\times10^{-19}} Joule$$

eV → electron Volt



$$\Delta E = E_4 - E_1$$

$$= \frac{-13.6}{4^2} - \frac{-13.6}{1^2} = 12.75\ eV$$



$h = 6.625\times10^{-34}\ J.S \qquad C = 3\times10^{8}\ m/s$

$$m = \frac{E(J)}{c^2} = \frac{12.75\times1.6\times10^{-19}}{[3\times10^8]^2} = \checkmark\ kg$$

$$\upsilon = \frac{E(J)}{h} = \frac{12.75\times1.6\times10^{-19}}{6.625\times10^{-34}} = \checkmark\ Hz$$

$$\lambda = \frac{hc}{E(J)} = \frac{6.625\times10^{-34}\times3\times10^{8}}{12.75\times1.6\times10^{-19}} = \checkmark\ m$$

$$P_L = \frac{E}{c} = \frac{12.75\times1.6\times10^{-19}}{3\times10^{8}} = \checkmark\ Kg.m/s$$

Hydroge Atom

∞ ionization theory

Q
P
O
N
M
L
K

Long IR $\lambda\uparrow \upsilon\downarrow E\downarrow$
Pfund's seires
n=5

Modrate IR
Brackett's seires
n=4

Short IR
Paschen's seires
n=3

Visible Light
Balmer's seires
n=2

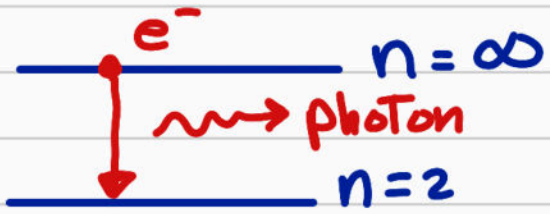
UV
Lyman's seires
n=1

$E_{UV} > E_{VL} > E_{IR}$
$\upsilon_{UV} > \upsilon_{VL} > \upsilon_{IR}$
$\lambda_{UV} < \lambda_{VL} < \lambda_{IR}$

nucleus

$E\uparrow \upsilon\uparrow \lambda\downarrow$	$E\downarrow \upsilon\downarrow \lambda\uparrow$
$E_\infty - E_n$	$E_{n+1} - E_n$
Gap أكبر فرق	Gap أصغر فرق

Find the highest Energy from Balmer's seires?



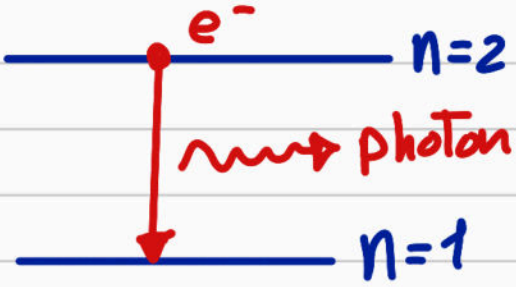
$$\Delta E = E_\infty - E_2$$

$$= \frac{-13.6}{\infty^2} - \frac{-13.6}{2^2}$$

$$= 0 - \frac{-13.6}{4} = \frac{13.6}{4}\ eV \xrightarrow{\times 1.6\times10^{-19}} Joule$$

The longest wavelength (λ) in Lyman seires

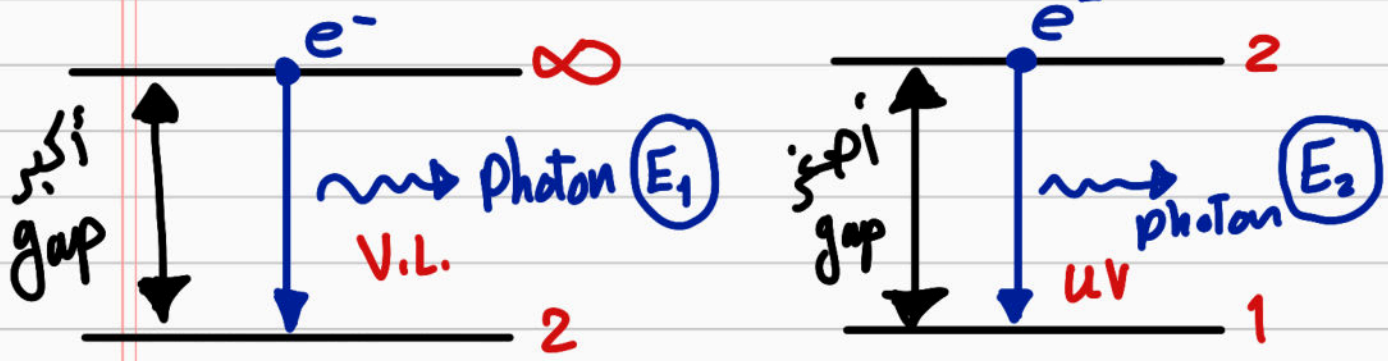
$\lambda\uparrow$ $E\downarrow$



$$\Delta E = E_2 - E_1 = \frac{-13.6}{2^2} - \frac{-13.6}{1^2}$$

$$= 10.2\ eV \xrightarrow{\times 1.6\times10^{-19}} Joule$$

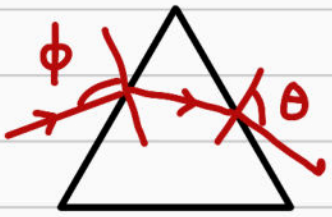
$$\lambda = \frac{hc}{\Delta E} = \frac{6.625\times10^{-34}\times 3\times10^{8}}{10.2\times1.6\times10^{-19}}$$



Visible photon أقوى VL $E_1 < E_2$ UV photon أضعف

3 Spectrometer / Spectroscope

$\phi = \theta$



#Sci. Idea: Min. deviasion angle (α_o)

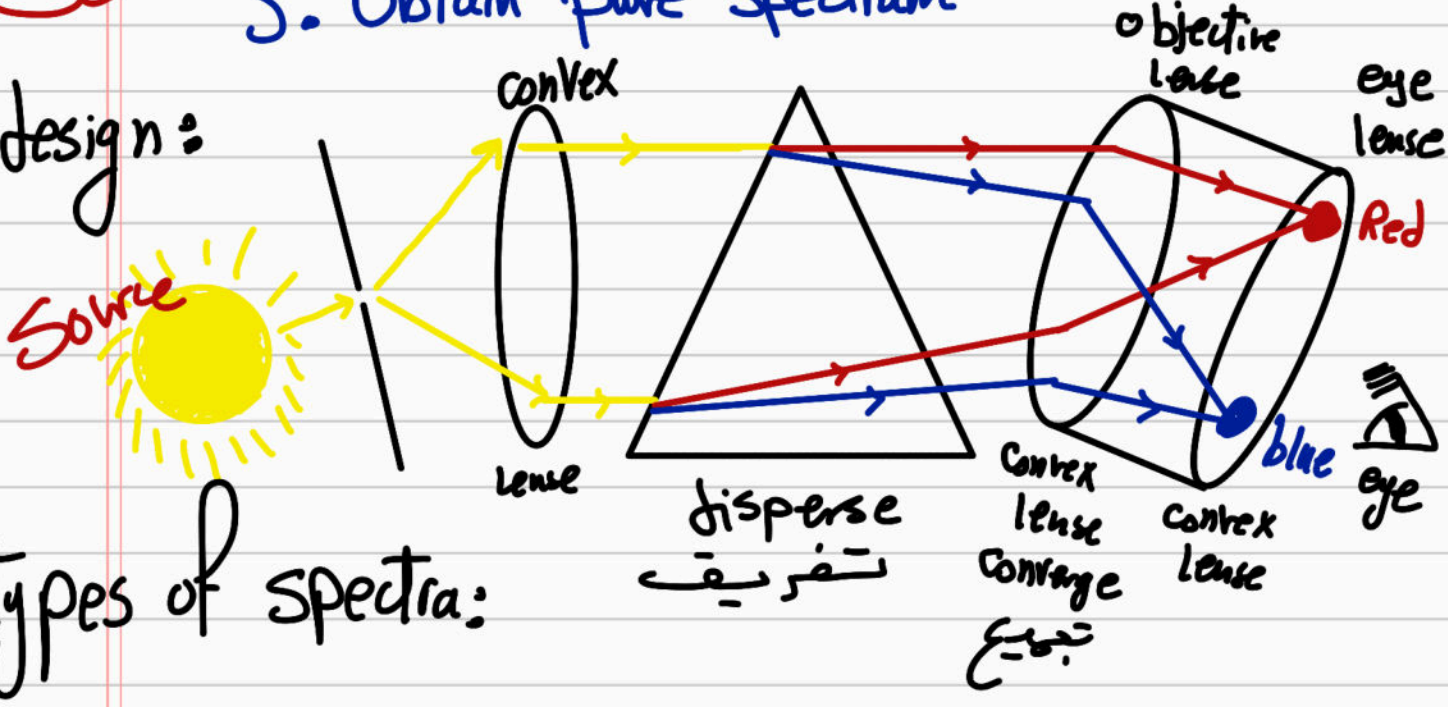
#Uses: 1. analysis light into Visible & inVisible

$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$

2. Estimate Temp. of stars & gases

3. Obtain Pure spectrum

#design:



#types of spectra:

1 Continouse Emission

Tungestn bulb





2 Line Absorption



Tungestn bulb

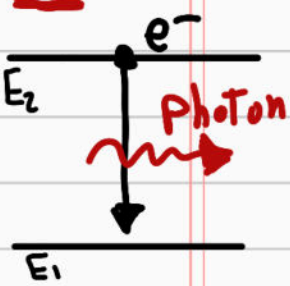




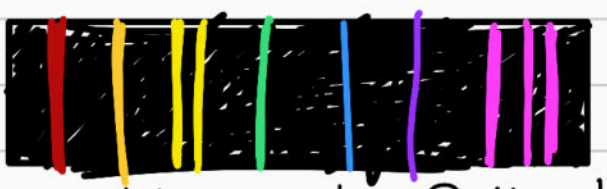
dark Lines on Colored background

3 Line Emission

same Gas



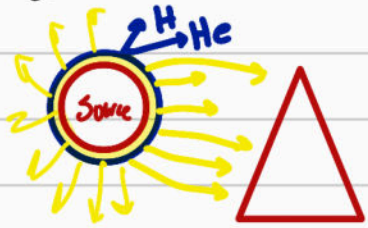




Colored Lines on dark Background

FranhoVer ✓✓

"Franhofer lines"





الغلاف الخارجي

The outer envelope of the Sun → Hydrogen H_1, Heliume He_2

2026

Colored Lines on dark background



Line Emission

different gases



Line Absorption

Dark lines on colored background

X-rays

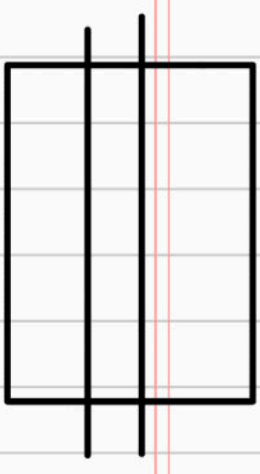
- properties
- Coolidge Tube
- Types
  - Cont.
  - Line

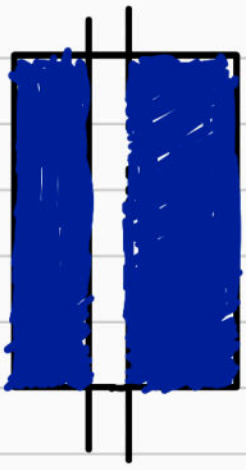
"X-Rays"

1 Properties:

IR	V.L.	UV	X	γ
7×10^{-7} m	4×10^{-7} m	10^{-8} m	10^{-13} m	
700nm	400nm	10nm	0.0001nm	
7000 A°	4000 A°	100 A°	0.001 A°	

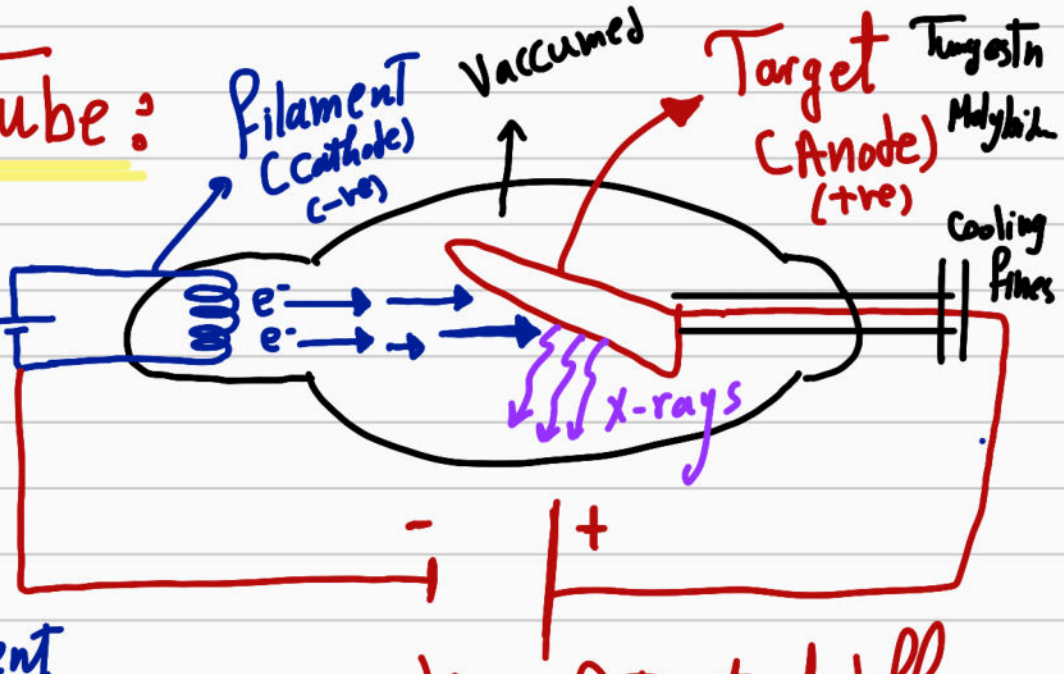
* high energy → Can ionize inert gases → He_2, Ne_{10}
* Short Wavelength ($\lambda \downarrow$) → high Penetrating Power ($P_{w_{pen}} \uparrow$)
  → Bone imaging
* *Diffract Crystals: → detect Cracks & Weldings
* *Chemical effect: → affect sensitive photographic Plates







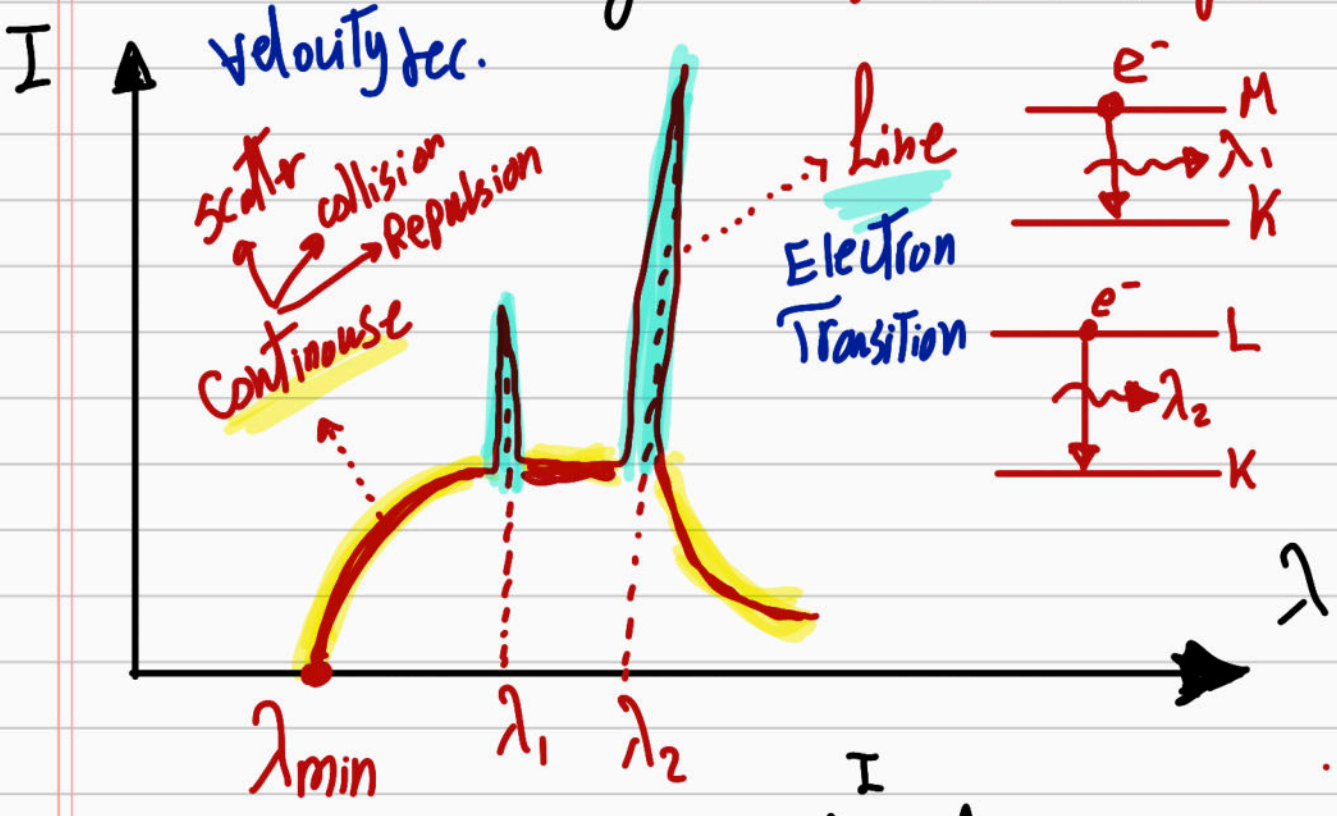
2 Coolidge Tube:

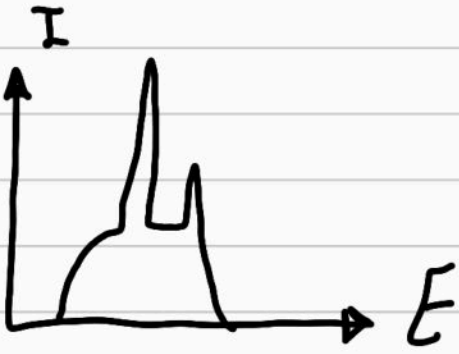


Filament Voltage → V_1

Electric → Thermal Filament → Mechanical "KE" → ElectroMagnetic "X-rays"

V_2 → Potential diff. between Cathode & Anode
Filamat & Target





Continouse
Connected / soft / braking

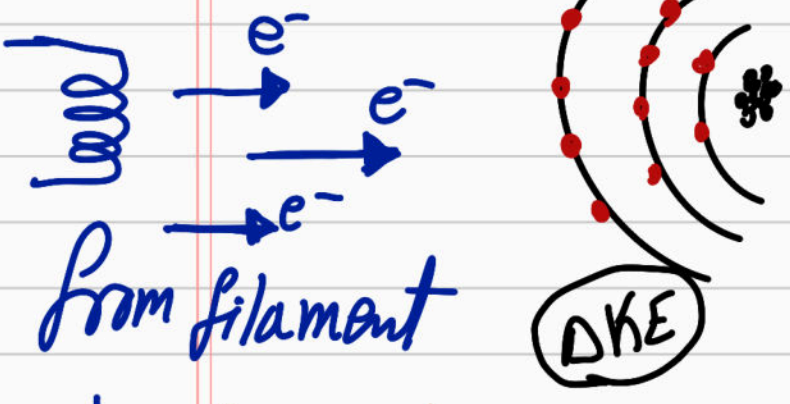
Line
Hard / Tough / Charachter

Scatter, Repulsion, Collision

→ Velocity decreases ←

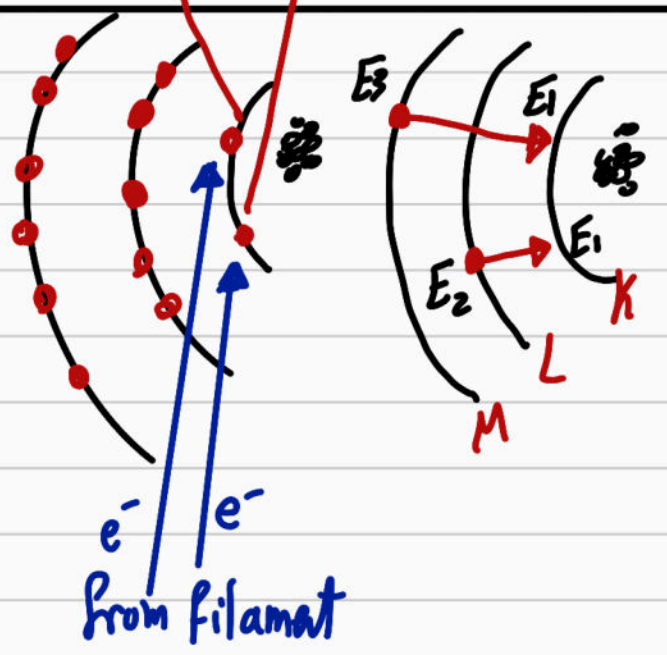
Electron Transition





from filament

↓ KE ⇝ X-rays



$$\lambda_{min} = \frac{hc}{eV_2} = \frac{hc}{KE_{max}}$$

filament → V_2

Target

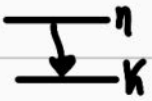
$$\lambda_{line} \propto \frac{1}{Z}$$

↓ λ ↑ Z

atomic no. of Target material

electrons from filament
(deceleration) ↓↓

electrons from Target
(Transition)



$$\lambda_{cont.\ min} \propto \frac{1}{V_2}$$

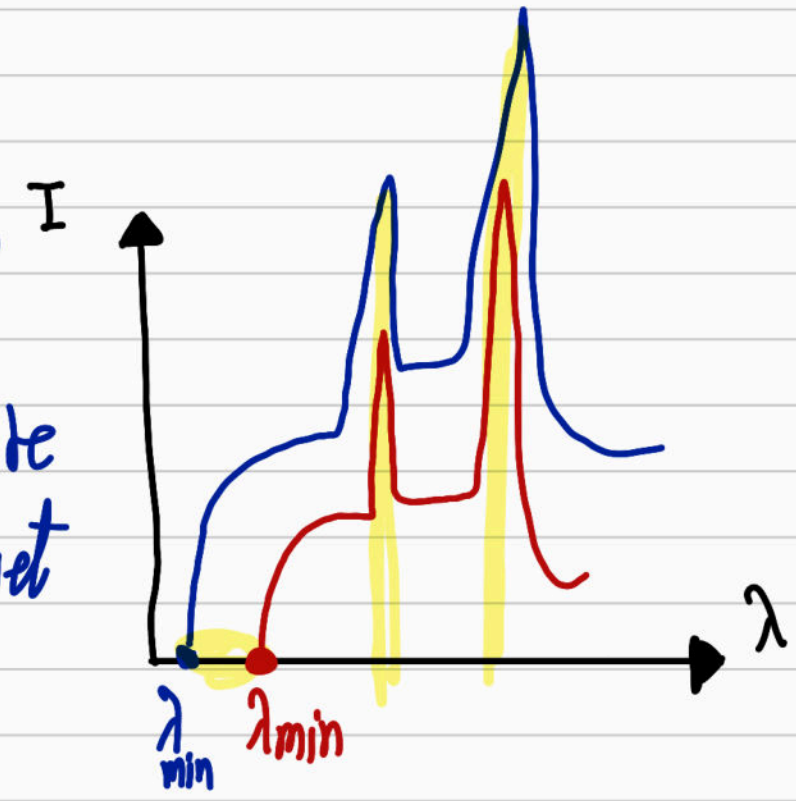
V_2 Cathode & Anode

$\lambda_{line} \propto \frac{1}{\text{Atomic no. Target}}$

V_1 ∝ I ↑

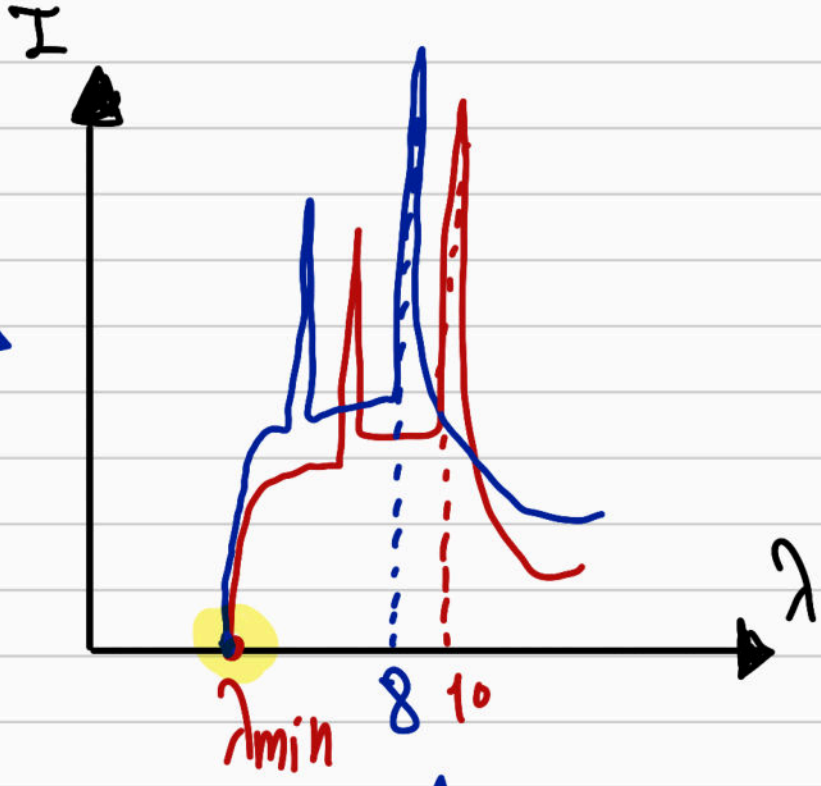
$\downarrow \lambda_{min} \propto \frac{1}{V_2 \uparrow}$

increasing potential diff. between cathode & Anode filament & Target

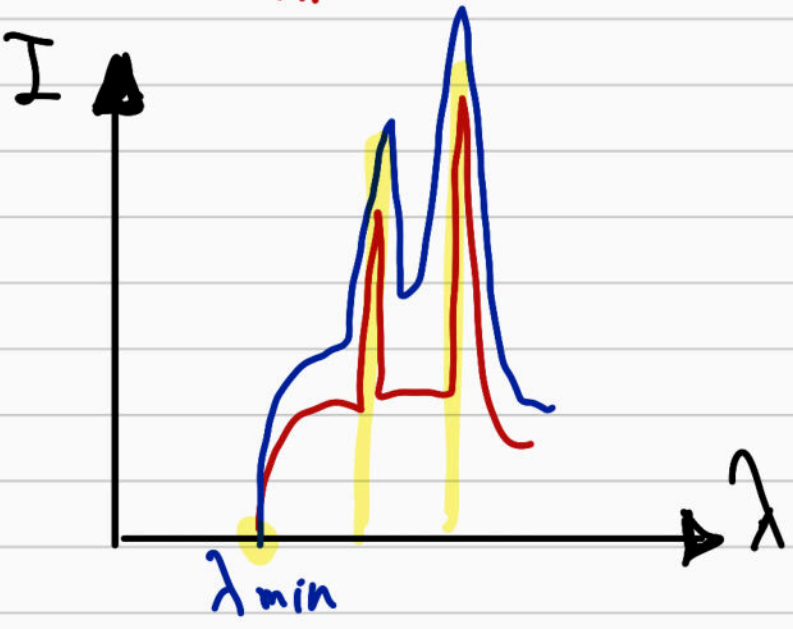


$\downarrow \lambda_{Line} \propto \frac{1}{\text{atomic no. Target} \uparrow}$

increasing the atomic no. of Target material



increasing filament Voltage/intesity (V_1)



P.O.C.	λ_{min}	λ line	I
increasing filament intensity	No change	No change	increase
decreasing potential diff. Cathode & Anode	increase	No change	decrease
increasing atomic no. of Target material	No change	decrease	increase